

Quiz Makeup

Q1. A university database maintains student records. Two transactions are running at the same time:

- ★ **Transaction T1 (Faculty):** Updates and checks student grades.
- ★ **Transaction T2 (Admin):** Calculates reports of average grades in the department.

- T1 reads Alice's grade = 80.
- T2 also reads Alice's grade = 80.
- T1 updates Alice's grade to 90 (but has not committed yet).
- T2 calculates the average grade including Alice's temporary grade = 90.
- T1 crashes and rolls back, Alice's grade remains 80.
- T2 still used the uncommitted value 90.
- Later, T1 again checks all "CSE" students where grade > 85.
- In the first query, it finds 2 students.
- Meanwhile, T2 inserts a new student Bob with grade 95.
- T1 repeats the same query and now finds 3 students.

- a. Identify the concurrency problems in the scenario. [4]
- b. State which ACID property is being violated for each problem. [4]
- c. How can this issue be prevented? (hint: mention isolation level). [2]

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- a. **Incorrect Summary:** value changed while average is calculated.
Dirty Read Problem: crashed and resets value, but other transactions uses it
Phantom Read: 1 student increased while checking info for students with grades above 85.
 - b. **Isolation, Consistency**
 - c. **Isolation level: serializable**

Q1. A bank database maintains customer account balances. Two transactions are running at the same time:

- **Transaction T1 (ATM Withdrawal):** Deducts money from Alice's account
- **Transaction T2 (Online Transfer):** Transfers money from Alice's account to Bob's account.

- T1 reads Alice's balance = 1000.
- T2 also reads Alice's balance = 1000.
- T1 deducts 200, sets temporary new balance = 800 (not committed yet).
- T2 deducts 300, sets temporary new balance = 700 (overwriting T1's update).
- T1 commits its balance = 800.
- Final balance in the database = 800 (T2's deduction is lost).
- Later, T1 queries Alice's balance = 800.
- Meanwhile, T2 deposits 500 into Alice's account and commits new balance = 1300.
- T1 queries Alice's balance again and now finds 1300.

- a. Identify the concurrency problems in the scenario. [4]
- b. State which ACID property is being violated for each problem. [4]
- c. How can this issue be prevented? (hint: mention isolation level). [2]

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- a. **Lost Update:** for T1 update being overwritten (lost)
Non-Repeatable Read: Alice sees balance is 1000 then 800.
 - b. **Isolation, Consistency**
 - c. **Isolation level: serializable**